

Day 5 Mastery Quiz: Boolean Combinations — Instructor Key

State/type/boundary evidence → and/or/not → readiness explanation

Instructor key. Same layout as student copy; solutions filled in response boxes. Print duplex on one sheet.

Name		Section		Date	
Score	_____/10		Mastery check		secure / developing / needs support

The pump team is preparing a final readiness note before the larger Week 1 closeout event. The code below combines current state, typed values, boundary evidence, and approval/setup evidence. Decide the Boolean values and explain which condition controls the result. Do not write an if statement.

1 Reference code for Items 1–10

trace debug

```
1 # converted values and setup evidence
2 flow_text = "21.6"
3 flow_lpm = float(flow_text)
4 low_limit_lpm = 20.4
5 high_limit_lpm = 21.6
6 approval_label = "reviewed"
7 operator_initials = ""
8 retry_count = 0
9 sensor_ready = True
10
11 # separate evidence checks
12 lower_ok = flow_lpm >= low_limit_lpm
13 upper_ok = flow_lpm <= high_limit_lpm
14 flow_ok = lower_ok and upper_ok
15 approval_ok = approval_label == "reviewed"
16 setup_ok = sensor_ready and operator_initials != ""
17 no_retry_needed = retry_count == 0
18 record_ready = flow_ok and approval_ok and setup_ok and no_retry_needed
19 needs_followup = (not flow_ok) or (not approval_ok) or (not setup_ok)
```

2 Items 1–5: trace the separate evidence

trace predict

Pt.	Prompt	My response
1	What values are stored in <code>lower_ok</code> and <code>upper_ok</code> ? Explain the exact upper boundary.	Key: Both are <code>True</code> ; <code>21.6 <= 21.6</code> passes because the upper boundary is inclusive.
1	What value is stored in <code>flow_ok</code> , and which connector creates it?	Key: <code>True</code> , created by <code>lower_ok</code> and <code>upper_ok</code> .
1	What value is stored in <code>approval_ok</code> ? Explain why this is not a truthiness check.	Key: <code>True</code> ; it uses explicit equality <code>approval_label == "reviewed"</code> , not non-empty-string truthiness.
1	What value is stored in <code>setup_ok</code> , and which part controls it?	Key: <code>False</code> ; <code>operator_initials != ""</code> is false because initials are blank.
1	What value is stored in <code>no_retry_needed</code> ?	Key: <code>True</code> because <code>retry_count == 0</code> .

3 Items 6–8: combined readiness and follow-up

[debug](#)
[explain](#)

Pt.	Prompt	My response
1	What value is stored in <code>record_ready</code> , and which one required condition prevents readiness?	Key: False; <code>setup_ok</code> is false.
1	What value is stored in <code>needs_followup</code> , and which <code>not(...)</code> term makes it true?	Key: True; <code>not setup_ok</code> is true.
1	If <code>operator_initials</code> changed to "AM" and the evidence were recomputed, what would happen to <code>setup_ok</code> and <code>record_ready</code> ?	Key: <code>setup_ok</code> would become True, and with the other checks true, <code>record_ready</code> would become True.

4 Item 9: debug an unsafe readiness claim

[debug](#)
[test](#)

Readiness note to check

The record is ready because the flow is in range or the approval label is reviewed.

Pt.	Prompt	My response
1	Explain why the claim uses the wrong connector and misses one required setup condition.	Key: It uses <code>or</code> , but readiness requires all checks with <code>and</code> . It also ignores blank <code>operator_initials</code> , so setup fails.

5 Item 10: write the correct combined readiness expression

[implement](#)
[transfer](#)

Pt.	Prompt	My response
1	Write the correct expression for readiness using all four required evidence checks and <code>and</code> .	Key: <code>flow_ok and approval_ok and setup_ok and no_retry_needed</code> .

Support route, not scored

My issue is mostly: setup / Python syntax / state history / type conversion / boundary checks / Boolean connector / `not` logic / confidence / time.
 My next small action is: list each evidence value / circle the first false required condition / test one failed condition at a time / ask a partner / ask instructor.